

```
mysql> select * from customers;
```

customer_id	customer_name	city	customer_type
1	Rahul	Delhi	Premium
2	Meena	Mumbai	Regular
3	Arjun	Bangalore	Premium
4	Kavya	Delhi	Regular
5	Suresh	Chennai	Premium
6	Anjali	Mumbai	Regular
7	Vikram	Delhi	Regular
8	Pooja	Bangalore	Premium
9	Devi	Chennai	Regular

```
9 rows in set (0.01 sec)
```

```
mysql> select * from products;
```

product_id	product_name	category	price
1	Laptop	Electronics	55000.00
2	Mobile	Electronics	25000.00
3	Chair	Furniture	7000.00
4	Table	Furniture	9000.00
5	Headphones	Electronics	3000.00
6	Bookshelf	Furniture	6000.00

```
6 rows in set (0.01 sec)
```

```
mysql> select * from orders;
```

order_id	customer_id	product_id	quantity	order_date	amount
1	1	1	1	2023-01-05	55000.00
2	2	2	1	2023-01-10	25000.00
3	3	3	2	2023-02-01	14000.00
4	1	5	2	2023-02-15	6000.00
5	4	4	1	2023-03-03	9000.00
6	5	2	2	2023-03-20	50000.00
7	6	4	3	2023-04-02	18000.00
8	3	5	1	2023-04-15	3000.00
9	7	3	1	2023-05-01	7000.00
10	8	1	1	2023-05-20	54000.00
11	2	3	2	2023-06-02	12000.00
12	5	4	1	2023-06-15	8500.00

```
12 rows in set (0.01 sec)
```

```
mysql> -- Multi-Row Subqueries ? ALL
```

```
mysql> -- Find products priced higher than ALL products in the 'Furniture'
```

category.

```
mysql> select product_name, price from products where price > all(select price from products where category='Furniture');
```

product_name	price
Laptop	55000.00
Mobile	25000.00

2 rows in set (0.00 sec)

```
mysql> select product_name, price from products where price < all(select price from products where category='Furniture');
```

product_name	price
Headphones	3000.00

1 row in set (0.00 sec)

```
mysql> -- Find orders with an amount greater than ALL orders placed by 'Meena'.
```

```
mysql> select order_id, amount from orders where amount > all(  
-> select amount from orders where customer_id=(select customer_id from customers where customer_name='meena'));
```

order_id	amount
1	55000.00
6	50000.00
10	54000.00

3 rows in set (0.00 sec)

```
mysql> -- Find the name(s) of the customer(s) who ordered the most expensive product.
```

```
mysql> select customer_name from customers where customer_id in(select customer_id from orders where product_id=(select product_id from products where price=(select max(price) from products)));
```

customer_name
Rahul
Pooja

2 rows in set (0.00 sec)

```
mysql>
```

```
mysql> -- Find the category of the product bought in the highest-amount order.
```

```
mysql> select category from products where product_id=(  
-> select product_id from orders
```

```
-> where amount=(select max(amount) from orders)
-> );
```

```
+-----+
| category |
+-----+
| Electronics |
+-----+
```

1 row in set (0.00 sec)

mysql> -- Find the city (or cities) of the customer(s) who bought the cheapest product.

```
mysql> select city from customers where customer_id in(
-> select customer_id from orders
-> where product_id =(
-> select product_id from products
-> where price=(select min(price) from products)
-> ));
```

```
+-----+
| city |
+-----+
| Delhi |
| Bangalore |
+-----+
```

2 rows in set (0.00 sec)

mysql> select customer_name from customers c where (select count(*) from orders o where o.customer_id =c.customer_id) > 1;

```
+-----+
| customer_name |
+-----+
| Rahul |
| Meena |
| Arjun |
| Suresh |
+-----+
```

4 rows in set (0.00 sec)

```
mysql> select p1.product_name,p1.category,p1.price
-> from products p1
-> where 1=(
-> select count(*) from products p2
-> where p2.category=p1.category and p2.price<p1.price
-> );
```

```
+-----+-----+-----+
| product_name | category | price |
+-----+-----+-----+
| Mobile | Electronics | 25000.00 |
| Chair | Furniture | 7000.00 |
+-----+-----+-----+
```

2 rows in set (0.00 sec)

```
mysql> select customer_name from customers c where exists (select 1 from orders o
where o.customer_id=c.customer_id);
```

```
+-----+
| customer_name |
+-----+
| Rahul         |
| Meena         |
| Arjun         |
| Kavya         |
| Suresh        |
| Anjali        |
| Vikram        |
| Pooja         |
+-----+
```

8 rows in set (0.00 sec)

```
mysql> select customer_name from customers c where not exists (select 1 from orders
o where o.customer_id=c.customer_id);
```

```
+-----+
| customer_name |
+-----+
| Devi          |
+-----+
```

1 row in set (0.00 sec)

```
mysql> select * from employees;
```

```
+-----+-----+-----+-----+
| emp_id | name   | department_id | salary |
+-----+-----+-----+-----+
| 101    | Arjun  | 1             | 50000.00 |
| 102    | Bhavana | 2             | 65000.00 |
| 103    | Charan | 2             | 72000.00 |
| 104    | Divya  | 3             | 45000.00 |
| 105    | Eswar  | NULL          | 40000.00 |
| 106    | Farhan | 4             | 55000.00 |
| 107    | Gowri  | 2             | 62000.00 |
| 108    | Harish | 5             | 48000.00 |
| 109    | Ishita | NULL          | 53000.00 |
| 110    | Jagan  | 3             | 60000.00 |
+-----+-----+-----+-----+
```

10 rows in set (0.02 sec)

```
mysql> select min(salary) as second_min_salary from employees
-> where
-> salary>(
-> select min(salary) as second_min_salary from employees);
```

```
+-----+
| second_min_salary |
+-----+
```

```
|          45000.00 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> select max(salary) as third_min_salary from employees
-> where salary<(
-> select max(salary) as second_max_salary from employees
-> where salary<(
-> select max(salary) as max_salary from employees));
```

```
+-----+
| third_min_salary |
+-----+
|          62000.00 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> select min(salary) as third_min_salary from employees
-> where salary>(
-> select min(salary) as second_min_salary from employees);
```

```
+-----+
| third_min_salary |
+-----+
|          45000.00 |
+-----+
1 row in set (0.00 sec)
```

```
mysql>
mysql>
mysql> select min(salary) as third_min_salary from employees
-> where salary>(
-> select min(salary) as second_min_salary from employees
-> where salary>(
-> select min(salary) as min_salary from employees));
```

```
+-----+
| third_min_salary |
+-----+
|          48000.00 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> select * from employees;
```

```
+-----+-----+-----+-----+
| emp_id | name   | department_id | salary |
+-----+-----+-----+-----+
| 101    | Arjun  | 1             | 50000.00 |
| 102    | Bhavana | 2             | 65000.00 |
| 103    | Charan | 2             | 72000.00 |
| 104    | Divya  | 3             | 45000.00 |
| 105    | Eswar  | NULL          | 40000.00 |
| 106    | Farhan | 4             | 55000.00 |
```

107	Gowri	2	62000.00
108	Harish	5	48000.00
109	Ishita	NULL	53000.00
110	Jagan	3	60000.00

10 rows in set (0.00 sec)

```
mysql> select * from departments;
```

department_id	department_name
1	Human Resources
2	Engineering
3	Marketing
4	Finance
5	Operations

5 rows in set (0.01 sec)

```
mysql> select d.department_id,d.department_name,min(e.salary) as second_min_salary
-> from employees e
-> join departments d
-> on e.department_id =d.department_id
-> where e.salary>(
-> select min(salary)
-> from employees e2
-> where e2.department_id=e.department_id)
-> group by d.department_id;
```

department_id	department_name	second_min_salary
2	Engineering	65000.00
3	Marketing	60000.00

2 rows in set (0.00 sec)

```
mysql> select d.department_id,d.department_name,max(e.salary) as second_max_salary
-> from employees e
-> join departments d
-> on e.department_id =d.department_id
-> where e.salary<(
-> select max(salary)
-> from employees e2
-> where e2.department_id=e.department_id)
-> group by d.department_id;
```

department_id	department_name	second_max_salary
2	Engineering	65000.00
3	Marketing	45000.00

```
+-----+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql>
```

```
mysql>
```

```
mysql>
```

```
mysql> /* Views
```

```
/*> a view is a virtual table created from the result of select query.
```

```
/*> it doesn't normally stores the actual data separately insted it stores the
SQL Query and whenever you accesses the view,
```

```
/*> my sql retrieves the data from the uderlying tables.
```

```
/*>
```

```
/*> Why Do We Need Views
```

```
/*> -----
```

```
/*> 1.Simplify complex queries
```

```
/*> 2.Reuse query logic:
```

```
/*> 3.Data security
```

```
/*> 4.Simplified reporting
```

```
/*>
```

```
/*> Advantages of Views
```

```
/*> -----
```

```
/*> 1. Security
```

```
/*> 2. Simplicity
```

```
/*> 3. Reusability
```

```
/*> 4.Consistency:
```

```
/*> 5.Maintainability
```

```
/*> 6. No extra storage cost */
```

```
mysql>
```

```
mysql> /* Diff b/w table and view
```

```
/*> -----
```

```
/*> table physically stores data
```

```
/*> view is virtual table based on query.
```

```
/*>
```

```
/*> can stores records directly
```

```
/*> usually doesn't store seperate records.
```

```
/*>
```

```
/*>
```

```
/*> create using create table
```

```
/*> create using create view.
```

```
/*>
```

```
/*> it contain all table data.
```

```
/*> can show selected rows/columns
```

```
/*>
```

```
/*>
```

```
/*> used for permanent data,storage.
```

```
/*> mainly used for abstract,security, and reusable queries.*/
```

```
mysql>
```

```
mysql> /* how to create the view
```

```
/*> -----
```

```
/*> create view syntax:
```

```

/*> -----
/*> create view view_name as select column from table where condition..
/*>
/*>
/*> altering view
/*> -----
/*> alter view view_nam as select new_column from new_tables where
new_condition;
/*>
/*>
/*> drop a view
/*> -----
/*> drop view view_name;
/*>
/*> */

```

```

mysql> drop table employees;
Query OK, 0 rows affected (0.04 sec)

```

```

mysql> drop table departments;
Query OK, 0 rows affected (0.02 sec)

```

```

mysql> CREATE TABLE departments (
-> dept_id INT PRIMARY KEY AUTO_INCREMENT,
-> dept_name VARCHAR(50),
-> location VARCHAR(50)
-> );
Query OK, 0 rows affected (0.03 sec)

```

```

mysql> CREATE TABLE employees (
-> emp_id INT PRIMARY KEY AUTO_INCREMENT,
-> emp_name VARCHAR(100),
-> dept_id INT,
-> designation VARCHAR(50),
-> salary DECIMAL(10,2),
-> join_date DATE,
-> FOREIGN KEY (dept_id) REFERENCES departments(dept_id)
-> );
Query OK, 0 rows affected (0.05 sec)

```

```

mysql> INSERT INTO departments (dept_name, location) VALUES
-> ('IT','Hyderabad'), ('HR','Bangalore'), ('Finance','Chennai'),
('Sales','Mumbai');
Query OK, 4 rows affected (0.01 sec)
Records: 4 Duplicates: 0 Warnings: 0

```

```

mysql>
mysql> INSERT INTO employees (emp_name, dept_id, designation, salary, join_date)
VALUES
-> ('Rahul',1,'Software Engineer',55000,'2021-06-15'),
-> ('Anita',2,'HR Executive',38000,'2020-03-21'),

```

```

-> ('Suresh',4,'Sales Manager',52000,'2019-08-10'),
-> ('Priya',1,'Team Lead',65000,'2022-01-05'),
-> ('Kiran',3,'Finance Analyst',48000,'2018-12-11'),
-> ('Deepika',2,'HR Manager',60000,'2019-10-19');

```

Query OK, 6 rows affected (0.01 sec)

Records: 6 Duplicates: 0 Warnings: 0

```
mysql> select * from employees;
```

emp_id	emp_name	dept_id	designation	salary	join_date
1	Rahul	1	Software Engineer	55000.00	2021-06-15
2	Anita	2	HR Executive	38000.00	2020-03-21
3	Suresh	4	Sales Manager	52000.00	2019-08-10
4	Priya	1	Team Lead	65000.00	2022-01-05
5	Kiran	3	Finance Analyst	48000.00	2018-12-11
6	Deepika	2	HR Manager	60000.00	2019-10-19

6 rows in set (0.00 sec)

```
mysql> select * from departments;
```

dept_id	dept_name	location
1	IT	Hyderabad
2	HR	Bangalore
3	Finance	Chennai
4	Sales	Mumbai

4 rows in set (0.00 sec)

```
mysql> create view vw_employee_department as
```

```

-> select e.emp_id,e.emp_name,e.designation,d.dept_name,d.location

```

```

-> from employees e

```

```

-> join departments d on e.dept_id=d.dept_id;

```

Query OK, 0 rows affected (0.02 sec)

```
mysql> show full tables where table_type='VIEW';
```

Tables_in_cg_mysql	Table_type
vw_employee_department	VIEW

1 row in set (0.00 sec)

```
mysql> select * from vw_employee_department;
```

emp_id	emp_name	designation	dept_name	location
1	Rahul	Software Engineer	IT	Hyderabad

4	Priya	Team Lead	IT	Hyderabad
2	Anita	HR Executive	HR	Bangalore
6	Deepika	HR Manager	HR	Bangalore
5	Kiran	Finance Analyst	Finance	Chennai
3	Suresh	Sales Manager	Sales	Mumbai

6 rows in set (0.00 sec)

```
mysql> create view vw_department_salary_summary as
-> select d.dept_name,
-> count(e.emp_id)as total_emp,
-> sum(e.salary)as total_sal,
-> avg(e.salary)as avg_sal
-> from employees e
-> join departments d on e.dept_id=d.dept_id
-> group by d.dept_name;
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> show full tables where table_type='VIEW';
```

Tables_in_cg_mysql	Table_type
vw_department_salary_summary	VIEW
vw_employee_department	VIEW

2 rows in set (0.00 sec)

```
mysql> select * from vw_employee_department;
```

emp_id	emp_name	designation	dept_name	location
1	Rahul	Software Engineer	IT	Hyderabad
4	Priya	Team Lead	IT	Hyderabad
2	Anita	HR Executive	HR	Bangalore
6	Deepika	HR Manager	HR	Bangalore
5	Kiran	Finance Analyst	Finance	Chennai
3	Suresh	Sales Manager	Sales	Mumbai

6 rows in set (0.00 sec)

```
mysql> select * from vw_department_salary_summary;
```

dept_name	total_emp	total_sal	avg_sal
IT	2	120000.00	60000.000000
HR	2	98000.00	49000.000000
Finance	1	48000.00	48000.000000
Sales	1	52000.00	52000.000000

4 rows in set (0.01 sec)

```
mysql> create view vw_employee_annual_salary as
-> select emp_name,salary as monthly_salary,salary*12 as anual_salary
-> from employees;
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> select * from vw_employee_annual_salary;
```

emp_name	monthly_salary	anual_salary
Rahul	55000.00	660000.00
Anita	38000.00	456000.00
Suresh	52000.00	624000.00
Priya	65000.00	780000.00
Kiran	48000.00	576000.00
Deepika	60000.00	720000.00

6 rows in set (0.00 sec)

```
mysql> create vw_high_paying_department A
```

```
-> ^C
```

```
mysql> create view vw_high_paying_department as
```

```
-> select dept_name,avg_sal
-> from vw_department_salary_summary
-> where avg_sal>50000;
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> select * from vw_high_paying_department;
```

dept_name	avg_sal
IT	60000.000000
Sales	52000.000000

2 rows in set (0.00 sec)

```
mysql> select * from vw_employee_department;
```

emp_id	emp_name	designation	dept_name	location
1	Rahul	Software Engineer	IT	Hyderabad
4	Priya	Team Lead	IT	Hyderabad
2	Anita	HR Executive	HR	Bangalore
6	Deepika	HR Manager	HR	Bangalore
5	Kiran	Finance Analyst	Finance	Chennai
3	Suresh	Sales Manager	Sales	Mumbai

6 rows in set (0.00 sec)

```
mysql> alter view vw_employee_department as
```

```

-> select e.emp_id,e.emp_name,d.dept_name
-> from employees e
-> join departments d on e.dept_id=d.dept_id;
Query OK, 0 rows affected (0.01 sec)

```

```
mysql> select * from vw_employee_annual_salary;
```

emp_name	monthly_salary	anual_salary
Rahul	55000.00	660000.00
Anita	38000.00	456000.00
Suresh	52000.00	624000.00
Priya	65000.00	780000.00
Kiran	48000.00	576000.00
Deepika	60000.00	720000.00

```
6 rows in set (0.00 sec)
```

```
mysql> select * from vw_employee_department;
```

emp_id	emp_name	dept_name
1	Rahul	IT
4	Priya	IT
2	Anita	HR
6	Deepika	HR
5	Kiran	Finance
3	Suresh	Sales

```
6 rows in set (0.00 sec)
```

```
mysql> show full tables where table_type='VIEW';
```

Tables_in_cg_mysql	Table_type
vw_department_salary_summary	VIEW
vw_employee_annual_salary	VIEW
vw_employee_department	VIEW
vw_high_paying_department	VIEW

```
4 rows in set (0.00 sec)
```

```
mysql> notee
```